

CLAS12 AI Tracking Validation Package

Package Architecture, Metrics, Histogram Variables, and User Guide

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Abstract

This note documents the `clas-ai-validation` package in the COATJAVA `common-tools` module. The package reads Monte Carlo truth associations, conventional and AI-assisted tracking banks, builds detector-neutral validation objects, matches reconstructed objects to generated particles, and produces object-level and particle-level performance histograms. Particular attention is given to the distinction between raw AI cluster suggestions, hit-based fitted objects, and time-based tracks. The note also documents every histogram written by `AITrackingPerformanceAnalysis`, the event-display controls, the histogram viewer, and command-line scripts for running all three applications.

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1 Purpose and scope

The package is a validation layer; it does not perform tracking. It observes banks produced by reconstruction services and converts them into a common data model. Its primary use is to compare:

- conventional hit-based and time-based DC reconstruction;
- raw AI cluster suggestions from `ai::tracks`;
- AI-assisted hit-based and time-based fitted outputs;
- reconstructed objects with MC truth at the hit, object, and particle levels.

The package supports event-by-event inspection through a Swing GUI and large-statistics studies through a GROOT histogram file.

2 High-level architecture

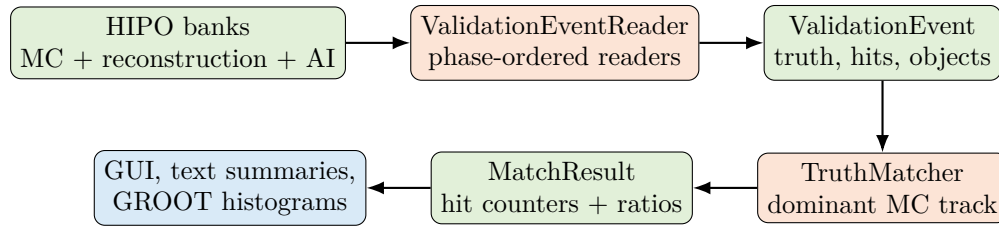


Figure 1: Validation data flow. Reconstruction is performed upstream; this package reads, matches, summarizes, and visualizes the resulting objects.

3 Package organization

Class or package	Responsibility
<code>data</code>	Validation objects: <code>Hit</code> , <code>Cluster</code> , <code>Segment</code> , <code>Cross</code> , <code>Seed</code> , <code>Track</code> , <code>AiTrackSuggestion</code> , <code>Particle</code> , and <code>ValidationEvent</code> .
<code>io.ValidationEventReader</code>	Runs all registered readers in truth, hit, cluster, segment, cross, seed, and track phases.
<code>io.mc.McTruthBankReader</code>	Reads <code>MC::Particle</code> and <code>MC::True</code> ; builds hit-to-truth associations.
<code>io.detector.DcBankReader</code>	Reads the available conventional and AI-assisted DC bank families while retaining their algorithm labels.
<code>io.detector.DcAiBankReader</code>	Reads raw AI cluster suggestions from <code>ai::tracks</code> , associates their referenced clusters and hits, and records the candidate probability.
<code>TruthIndex</code> and <code>TruthMatcher</code>	Provide event-local truth scopes and choose the dominant truth track for each validation object.

Class or package	Responsibility
DetectorMetrics and MatchResult	Store integer hit counters and derived purity, efficiency, and fake fraction.
service.AIValidationEngine	COATJAVA service wrapper and event-validation driver.
analysis.AITrackingPerformanceAnalysis	Fills event, stage, AI-score, and truth-efficiency histograms and writes a GROOT TDirectory HIPO file.
analysis.AITrackingHistogramViewer	Opens the histogram HIPO file in a GROOT TBrowser.
gui.TrackingValidationFrame	Standalone event browser with file selection, Previous/Next navigation, local X - Z and layer-wire views, and a metrics table.

4 Event-reading sequence

The event reader runs every registered detector reader once per phase. The phase ordering, rather than reader registration order within a phase, guarantees that truth and hit data exist before higher-level objects are matched.

1. **Truth phase:** generated particles and MC hit associations.
2. **Hit phase:** reconstructed detector hits and their truth owners.
3. **Cluster, segment, and cross phases:** intermediate tracking objects and their hit membership.
4. **Seed phase:** raw AI suggestions and hit-based fitted tracks represented as validation seeds.
5. **Track phase:** time-based fitted tracks.

A typical reader setup is:

```
ValidationEventReader reader = new ValidationEventReader();
reader.addReader(new McTruthBankReader());
reader.addReader(new DcBankReader());
reader.addReader(new DcAiBankReader());

ValidationEvent validationEvent = reader.readEvent(dataEvent);
```

5 Core data model

5.1 Measured-hit identity

A measured hit is identified by

$$\text{HitKey} = (\text{detector}, \text{sector}, \text{layer}, \text{component}, \text{hitId}). \quad (1)$$

For the DC, the bank stores a superlayer and a layer within that superlayer. The validation model uses the global layer

$$L_{\text{global}} = 6(SL - 1) + L_{\text{local}}, \quad 1 \leq L_{\text{global}} \leq 36. \quad (2)$$

The full key prevents accidental equality between hits from different sectors, layers, wires, or detectors even when a numerical hit ID is reused by another bank family.

5.2 Validation objects and object keys

Every reconstructed object supplies an ID, algorithm label, object type, detector scope, and a list of `HitKeys`. The object types are `CLUSTER`, `SEGMENT`, `CROSS`, `SUGGESTION`, `SEED`, and `TRACK`. The identity used by `ValidationEvent` is

$$(\text{algorithm}, \text{detector}, \text{type}, \text{id}). \quad (3)$$

The algorithm component is essential because conventional HB, conventional TB, AI-HB, and AI-TB banks may contain objects with the same numerical ID.

6 DC reconstruction families

Validation label	Typical source	Meaning
DC-HB	<code>HitBasedTrkg::HB*</code>	Conventional hit-based clusters, segments, crosses, and fitted HB tracks. HB tracks are represented as validation Seeds .
DC-TB	<code>TimeBasedTrkg::TB*</code>	Conventional time-based clusters, segments, crosses, and final tracks.
DC-AI-HB	<code>HitBasedTrkg::AI*</code> , or the standard HB family in an AI-only run	Hit-based output formed from AI-assisted pattern recognition.
DC-AI-TB	<code>TimeBasedTrkg::AI*</code> , or the standard TB family in an AI-only run	Time-based output formed from AI-assisted pattern recognition.
DC-AI-SUGGESTION	<code>ai::tracks</code>	Raw candidate containing a sector, a probability, and cluster IDs <code>c1</code> through <code>c6</code> ; it is not yet a fitted track.

The validation package keeps these families separate. A raw AI suggestion can have excellent hit recovery but later fail track fitting, while a final TB track can lose hits or acquire contamination during fitting. Combining these stages would hide where performance changes occur.

7 AI suggestion probability and hit score

Each row of `ai::tracks` has a candidate probability, denoted here by p_{AI} . The referenced clusters are marked as AI-suggested, and all hits belonging to those clusters are marked as AI-selected.

For DC, the package does not read an independently predicted score for every hit. Instead, the hit score is inherited from the best AI candidate that contains the hit's cluster:

$$s_{\text{hit}}(h) = \max_{a: h \in H(a)} p_{\text{AI}}(a). \quad (4)$$

Thus, several hits from the same candidate may have the same score. If a hit is contained in multiple candidates, `Hit.markAiSelected()` retains the largest finite candidate probability. The score is therefore a measure of the confidence of the best candidate selecting the hit, not a separately trained per-hit probability.

For CVT, the data model also permits a score assigned directly to an individual hit by a CVT AI algorithm.

8 Truth matching and metric definitions

8.1 Dominant truth track

For a reconstructed object O , the matcher counts the truth owner of every hit in $H(O)$ and selects the track with the largest overlap:

$$t^*(O) = \operatorname{argmax}_t |H(O) \cap H_t|. \quad (5)$$

An object with no truth-associated hits is unmatched.

8.2 Integer counters

For the selected truth track, `DetectorMetrics` stores:

Variable	Definition
<code>objectHits</code>	Number of reconstructed hits attached to the object.
<code>matchedHits</code>	Number of object hits associated with the selected truth track.
<code>truthHits</code>	Number of expected truth-associated hits in the comparison scope appropriate to the object.
<code>contaminatingHits</code>	$\max(0, \text{objectHits} - \text{matchedHits})$.
<code>missedHits</code>	$\max(0, \text{truthHits} - \text{matchedHits})$.

The ratios are computed from these counters:

$$P \equiv \text{purity} = \frac{N_{\text{matched}}}{N_{\text{object}}}, \quad (6)$$

$$\epsilon \equiv \text{efficiency} = \frac{N_{\text{matched}}}{N_{\text{truth}}}, \quad (7)$$

$$f_{\text{fake}} = \frac{N_{\text{object}} - N_{\text{matched}}}{N_{\text{object}}}. \quad (8)$$

A zero denominator produces a value of zero. For an object with a valid match, $f_{\text{fake}} = 1 - P$.

8.3 DC cluster truth scope

For a reconstructed DC cluster C matched to track t :

$$P(C) = \frac{\text{hits in } C \text{ from } t}{\text{all reconstructed hits in } C}, \quad (9)$$

$$\epsilon(C) = \frac{\text{hits in } C \text{ from } t}{\text{all truth-associated hits from } t \text{ in the same sector and superlayer}}. \quad (10)$$

For larger objects, the truth scope expands to the detector/sector regions represented by the object's hit set. The GUI reports the selected sector's truth hit count explicitly.

9 Histogram analysis workflow

`AITrackingPerformanceAnalysis` receives the already-built `ValidationEvent` and the list of `MatchResults` from `AIValidationEngine`. It does not reread banks and does not repeat truth matching. For each successful event it:

1. fills event-level occupancies;
2. fills per-hit AI score distributions;
3. fills one histogram group for every algorithm/object stage present;
4. fills raw AI-suggestion probability studies;
5. records which truth particles are recovered by each stage;
6. fills common truth denominators and stage-specific numerators.

The output is a GROOT TDirectory HIPO file with directories `/event`, `/truth`, `/ai`, and `/stages/<stage>`.

10 Complete histogram variable reference

10.1 Event-level histograms: `/event`

Histogram	Entries	Meaning
<code>event_hits</code>	One per event	Total measured DC hits stored in <code>ValidationEvent</code> . This is an event occupancy variable, not the number of hits on one track.
<code>event_truth_particles</code>	One per event	Number of generated particles loaded from <code>MC::Particle</code> .
<code>event_ai_suggestions</code>	One per event	Number of raw <code>AiTrackSuggestion</code> objects built from <code>ai::tracks</code> .
<code>event_seeds</code>	One per event	Number of validation seeds, including hit-based fitted tracks represented as seeds.

Histogram	Entries	Meaning
event_tracks	One per event	Number of time-based validation tracks.

10.2 Truth denominator histograms: /truth

A truth particle enters the momentum, theta, and phi denominators only when it has at least `minimumTruthHits` truth-associated DC hits. The default is five. This suppresses particles that do not traverse enough of the DC to form a meaningful reconstruction target.

Histogram	Variable	Meaning
truth_p_denominator	p in GeV	Generated momentum distribution of eligible truth particles.
truth_theta_denominator	θ in degrees	Generated polar-angle distribution, $\theta = \cos^{-1}(p_z/p)$.
truth_phi_denominator	ϕ in degrees	Generated azimuth, $\phi = \text{atan2}(p_y, p_x)$.
truth_dc_hits	DC hit count	Number of distinct truth-associated measured DC hits per truth track. This histogram is filled before the minimum-hit cut and shows the denominator-selection population.

10.3 Raw AI variables: /ai

Histogram	Type	Definition and interpretation
ai_probability_all	1D	Candidate probability for every finite-probability AI suggestion. One entry per suggestion.
ai_probability_matched	1D	Candidate probability for suggestions with at least one truth-associated hit and a selected dominant truth track.
ai_probability_unmatched	1D	Candidate probability for suggestions that receive no valid truth match.
ai_clusters_per_suggestion	1D	Number of positive cluster IDs among <code>c1...c6</code> that are attached to each suggestion.
ai_hits_per_suggestion	1D	Number of distinct reconstructed hits in the union of the suggestion's referenced clusters.
ai_purity_vs_probability	2D	$x = p_{\text{AI}}$, $y = P$ for each suggestion. Tests whether candidate probability is calibrated as a measure of hit purity.
ai_efficiency_vs_probability	2D	$x = p_{\text{AI}}$, $y = \epsilon$ for each suggestion. Tests whether high-probability candidates recover a larger fraction of the truth hits.
ai_hits_vs_probability	2D	$x = p_{\text{AI}}$, $y = N_{\text{object}}$. Reveals whether candidate probability depends on candidate size.

Histogram	Type	Definition and interpretation
ai_hit_score_all	1D	Best AI candidate score assigned to every AI-selected hit. One entry per selected hit, not per suggestion.
ai_hit_score_signal	1D	Hit scores for AI-selected hits with <code>truthTrackId > 0</code> . These are truth-associated signal hits, independent of which suggestion is the dominant object match.
ai_hit_score_background	1D	Hit scores for AI-selected hits with no truth-track association. This distribution is useful for score-threshold studies and background rejection.

10.4 Stage directories: /stages/<stage>

A stage label is formed as

$$\text{stageLabel} = \text{ALGORITHM/OBJECT_TYPE}. \quad (11)$$

The directory name is a lowercase, underscore-separated version. Examples are `dc_hb_cluster`, `dc_tb_track`, `dc_ai_hb_seed`, and `dc_ai_suggestion_suggestion`. Every stage that occurs in the input gets the following histograms.

Suffix	Type	Definition
purity	1D	Object purity $N{\text{matched}}/N_{\text{object}}$. The current analysis fills this for both matched and unmatched objects; unmatched objects therefore contribute at zero.
efficiency	1D	Object hit efficiency $N{\text{matched}}/N_{\text{truth}}$. Unmatched objects contribute at zero. This is not the particle-level tracking efficiency.
_fake_fraction	1D	Fraction of object hits not assigned to the selected truth track. Unmatched nonempty objects generally contribute near one.
_object_hits	1D	Total hits attached to every object, matched or unmatched. A zero entry means that an object with an empty hit list reached the analysis and should be investigated in the reader.
_matched_hits	1D	Truth-matched hit count. Filled only when <code>result.isMatched()</code> is true, so unmatched objects do not create an artificial zero peak here.
_truth_hits	1D	Expected truth-hit denominator in the object's truth scope. Filled for every object; unmatched objects have zero because there is no selected truth track.

Suffix	Type	Definition
<code>_purity_vs_efficiency</code>	2D	Object-level correlation between purity and hit efficiency.
<code>_matched_vs_truth_hits</code>	2D	$x = N_{\text{truth}}$, $y = N_{\text{matched}}$, filled only for matched objects. The diagonal is complete hit recovery; points below it represent missed truth hits.
<code>_objects_per_event</code>	1D	Total number of objects of this stage in each event in which the stage occurs.
<code>_matched_objects_per_event</code>	1D	Number of truth-matched objects of this stage per event.
<code>_unmatched_objects_per_event</code>	1D	Number of unmatched objects of this stage per event.
<code>_objects_per_truth_track</code>	1D	Number of matched objects from the stage assigned to the same truth track in one event. Values above one quantify clones or split reconstruction.
<code>_truth_p_numerator</code>	1D	Generated momentum of eligible truth particles recovered by at least one object in this stage. Each truth track is counted once per event and stage.
<code>_truth_theta_numerator</code>	1D	Generated theta of eligible truth particles recovered by the stage.
<code>_truth_phi_numerator</code>	1D	Generated phi of eligible truth particles recovered by the stage.

10.5 Object efficiency versus particle-level tracking efficiency

The histogram `<stage>_efficiency` is an object-level hit-recovery ratio. It answers: “For this reconstructed object and its selected truth particle, what fraction of the expected truth hits was recovered?”

Particle-level tracking efficiency is obtained by dividing a stage numerator by the common denominator bin-by-bin, for example

$$\epsilon_{\text{track}}^{(s)}(p) = \frac{N_{\text{recovered}}^{(s)}(p)}{N_{\text{eligible truth}}(p)}. \quad (12)$$

The same construction applies to theta and phi. A truth track contributes at most once to a stage numerator even if the stage produces multiple matching objects.

10.6 Understanding zero peaks

Zero entries can have several meanings and should not be interpreted uniformly:

- A zero in `_purity`, `_efficiency`, or `_truth_hits` may represent a legitimate unmatched object because those histograms are filled for all objects.

- A zero in `_matched_hits` should not arise from unmatched objects in the current implementation because that histogram is filled only for matched results.
- A zero in `_object_hits` indicates an object with no resolved hit membership and is generally a bank-reader or reference-resolution problem rather than a physical object.
- Empty `ai_hit_score_signal` or `ai_hit_score_background` histograms mean that no selected hits of that category had a finite score. They are not automatically evidence of a plotting failure.

11 Event-display GUI

The standalone frame opens a reconstructed HIPO file and caches validated events for Previous/Next navigation. The top controls select an MC track and a sector and toggle four overlaid hit selections:

- **MC truth:** open blue circles;
- **HB matched:** orange squares;
- **TB matched:** filled green circles;
- **AI-cluster matched:** purple diamonds/crosses.

The **local X–Z** view uses DC chamber-local X at the chamber midplane versus the corresponding midplane Z . Local X may be negative. The **layer–wire** view gives a topology-oriented representation independent of chamber coordinates.

The table below the plot is evaluated for the chosen MC track and sector. Its columns are object hits, truth-matched hits, truth hits in the selected sector, purity, and efficiency. The overlaid markers can coincide exactly; the table is therefore essential for distinguishing equal hit recovery from different contamination levels.

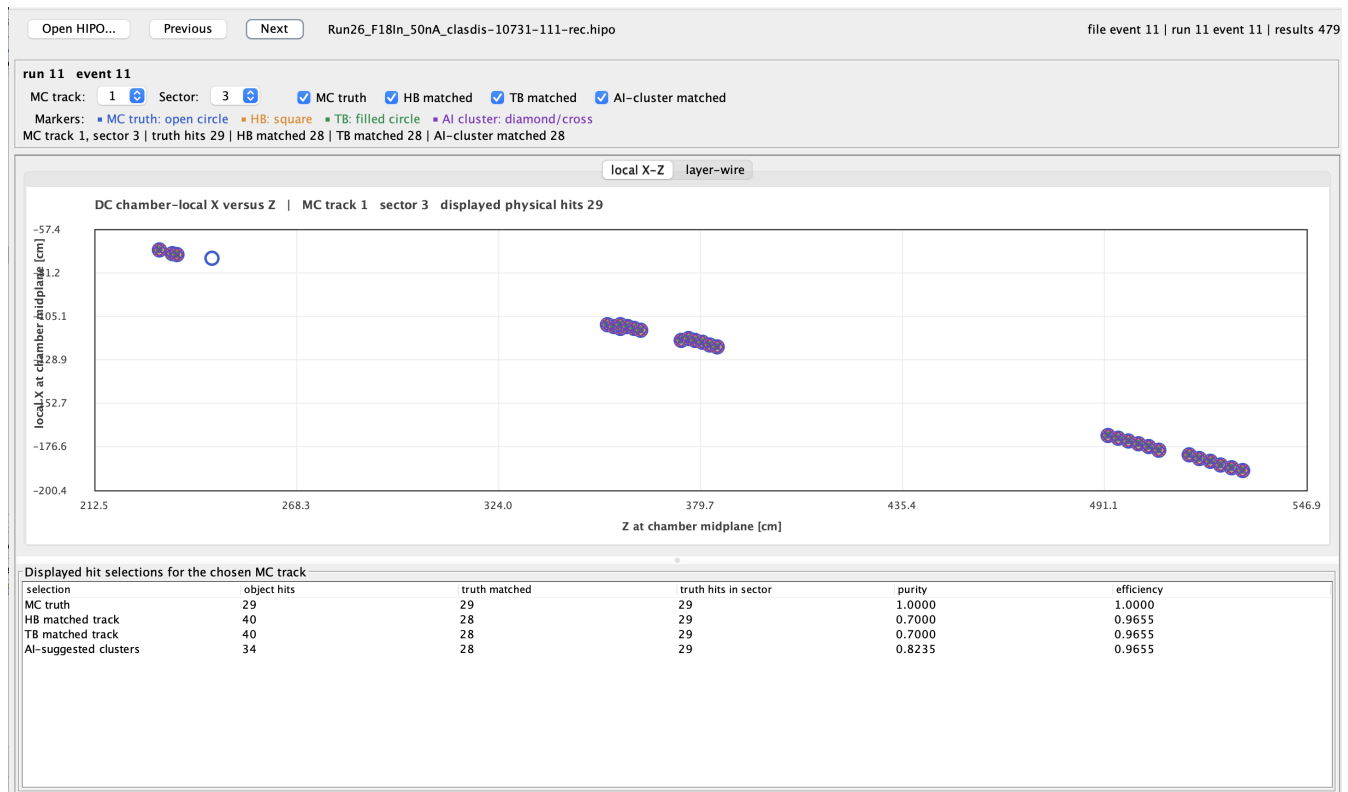


Figure 2: Event-display example with high truth-hit recovery. The selected MC track has 29 truth hits in sector 3. HB and TB recover 28 hits, while the AI-suggested clusters also recover 28 but contain fewer total hits, producing higher purity than the fitted HB/TB track in this event.

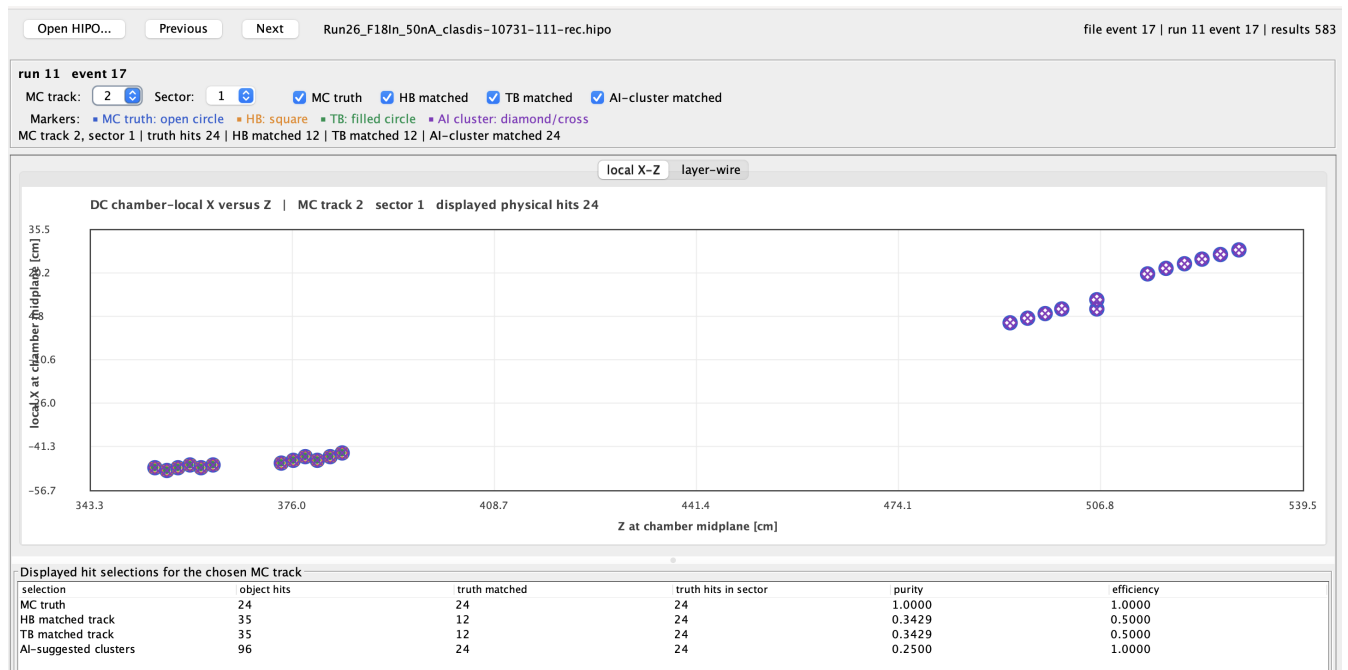


Figure 3: Event-display example illustrating why purity and efficiency must both be reported. The AI-suggested clusters recover all 24 truth hits (efficiency 1.0) but contain 96 hits in total (purity 0.25). HB and TB recover only half of the truth hits and also contain substantial contamination.

12 Histogram viewer

`AITrackingHistogramViewer` reads the GROOT HIPO histogram file and opens a `TBrowser`. The directory tree separates global AI variables, event occupancies, truth denominators, and stage-specific groups. Selecting a dataset renders it in the canvas; two-dimensional histograms can be displayed with a logarithmic color scale in the browser.

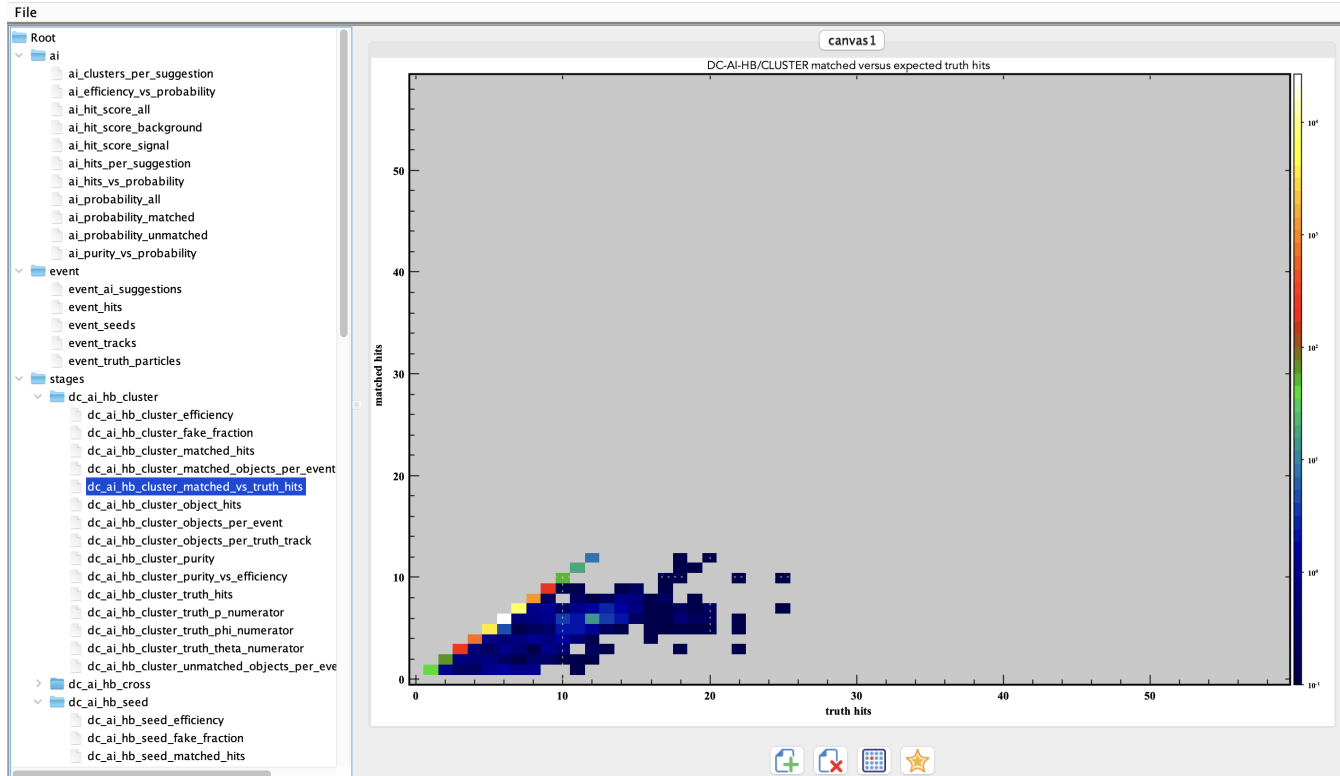


Figure 4: Histogram browser showing `dc_ai_hb_cluster_matched_vs_truth_hits`. The horizontal axis is the expected truth-hit count and the vertical axis is the number recovered by the cluster. The diagonal represents complete hit recovery.

13 Command-line operation

The supplied scripts are stored inside the `clas-ai-validation` module. All scripts use the Maven Exec Plugin by its fully qualified goal, so no edit to the module `pom.xml` is required.

13.1 Run the histogram analysis

```
./run_analysis.sh INPUT.hipo [OUTPUT.hipo] [MAX_EVENTS] [MIN_TRUTH_HITS]
```

Examples:

```
# Analyze all events, use the default output name, and require 5 truth hits.
./run_analysis.sh Run26_F18In_50nA_clasdis-10731-111-rec.hipo

# Analyze 100000 events and require at least 6 truth-associated DC hits.
```

```
./run_analysis.sh input.hipo results/ai_histos.hipo 100000 6
```

A nonpositive MAX_EVENTS means all events. When no output path is given, the Java analysis writes <input-base>_ai_validation_histos.hipo beside the input file.

13.2 Open the histogram viewer

```
./run_histogram_viewer.sh results/ai_histos.hipo
```

When no argument is supplied, the Java viewer opens a file chooser.

13.3 Open the event-display GUI

```
./run_tracking_gui.sh Run26_F18In_50nA_clasdis-10731-111-rec.hipo
```

When no argument is supplied, the frame opens a HIPO file chooser. The script launches `TrackingValidationFrame` directly, avoiding the older launcher's machine-specific hard-coded `CLAS12DIR` value.

14 Recommended validation checks

Before producing report plots, verify the following:

1. The analysis reports zero failed events, or each failed event is understood.
2. Stage `_object_hits` histograms do not have unexplained zero entries.
3. The number of raw AI suggestions in `event_ai_suggestions` agrees with the input bank for spot-checked events.
4. Event-display hit counts agree with the corresponding `MatchResult` counters.
5. Particle-efficiency numerators are divided by the matching common truth denominator with identical binning.
6. High AI probability is not treated automatically as high purity; inspect `ai_purity_vs_probability` and `ai_efficiency_vs_probability` directly.

15 Interpretation summary

The package provides three complementary levels of validation:

- **Hit level:** AI-selected hit scores, truth association, signal/background score distributions.
- **Object level:** purity, hit efficiency, fake fraction, hit counts, and clones for clusters, segments, crosses, suggestions, seeds, and tracks.
- **Particle level:** recovered truth-particle numerators over common generated-particle denominators versus momentum, theta, and phi.

No single metric is sufficient. A candidate may be fully efficient but highly contaminated, or very pure but incomplete. The event display makes these cases visually apparent, while the histogram analysis quantifies them over large data samples.